

ARASH JALIL KHABBAZI

✉ arashjkh@purdue.edu
📍 West Lafayette, IN, USA

🌐 ajkhabbazi.github.io
🌐 linkedin.com/in/ajkhabbazi

🎓 [Google Scholar](#)

SUMMARY

I'm a PhD candidate in Mechanical Engineering at Purdue University, working on modeling, control, optimization, and machine learning for building energy systems. My research builds practical automation for efficient, occupant- and grid-interactive buildings, focusing on predictive control and reinforcement learning. I develop and demonstrate scalable data-driven control algorithms in real buildings, and I also work on power control systems in homes and on demand flexibility in commercial buildings. My work combines rigorous methods with real-world testing to make control systems more reliable and easier to deploy at scale.

EDUCATION

Purdue University , <i>West Lafayette, IN</i> PhD in Mechanical Engineering Concentration in Computational Science and Engineering	<i>08/2023 – Present</i> GPA: 3.9/4.0
University of British Columbia (UBC) , <i>BC, Canada</i> MS in Mechanical Engineering <i>Thesis: "Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines"</i>	<i>05/2021 – 08/2023</i> GPA: 94% (4.0/4.0)
University of Tabriz , <i>EA, Iran</i> BS in Mechanical Engineering	<i>08/2016 – 07/2020</i> GPA: 19.12/20 (4.0/4.0) – Rank: 1/155+

SELECT AWARDS AND RECOGNITION

ASHRAE Graduate Student Grant-in-Aid Award	<i>2025</i>
Best PhD Forum Presentation Runner-up Award , <i>ACM BuildSys</i>	<i>2025</i>
ACM SIGEnergy Student Travel Grant , <i>ACM BuildSys</i>	<i>2025</i>
Best Paper Award , <i>CSME International Congress</i>	<i>2023</i>
Best Presentation Award , <i>CSME International Congress (Advanced Energy Symposium)</i>	<i>2022</i>
Graduate Research Scholarship , <i>University of British Columbia</i>	<i>2022</i>
Graduate Dean's Entrance Scholarship , <i>University of British Columbia</i>	<i>2021</i>

PUBLICATIONS

Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review." *Applied Energy*, 2025. 🔗
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions." *International Journal of Hydrogen Energy*, 2024. 🔗

Preprints

1. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher, "Indirect Data-Driven Predictive Control and the State-Space Predictor." *arXiv:2602.10936*, 2026. *Under review*.

Conference Papers

6. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher, "Scalable Supervisory HVAC Control for Linear Objectives." *65th IEEE Conference on Decision and Control (CDC)*, 2026.
5. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher, "Evaluating Power Control Strategies for Avoiding Residential Panel Upgrades in Cold Climates." *ACEEE Summer Study on Energy Efficiency in Buildings*, 2026.
4. **A.J. Khabbazi** and K.J. Kircher, "Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings." *American Control Conference (ACC)*, 2026.
3. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, "Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study." *CISBAT*, 2025.

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference, 2024*.
1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *Canadian Society for Mechanical Engineering (CSME) International Congress, 2023. Best Paper Award*.

Abstracts

3. **A.J. Khabbazi**, “PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2025. Best PhD Forum Presentation Runner-up Award*.
2. **A.J. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *Canadian Society for Mechanical Engineering (CSME) International Congress, 2022. Best Presentation Award*.
1. **A.J. Khabbazi**, R. Li, J. Quinn, and M. Hoorfar, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII), 2021*.

TALKS

8. “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.” *American Control Conference (ACC), New Orleans, LA, 2026*.
7. “Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *PhD Forum, 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), Golden, CO, 2025*.
6. “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference, Purdue University, West Lafayette, IN, 2024*.
5. “Field Demonstrations of Advanced Commercial HVAC Control.” *Intelligent Building Operations Workshop, Purdue University, West Lafayette, IN, 2024*.
4. “Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines.” *Herrick Graduate Seminar, Purdue University, West Lafayette, IN, 2023*.
3. “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *CSME International Congress, Université de Sherbrooke, QC, Canada, 2023*.
2. “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *CSME International Congress, University of Alberta, AB, Canada, 2022*.
1. “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII), 2021*.

RESEARCH EXPERIENCE

- Purdue University**, Graduate Research Assistant, Kircher Research Group, *West Lafayette, IN* 08/2023–Present
- Building supervisory control software for a live field demonstration of data-driven HVAC control in a campus building.
 - Co-developed a commissioning-free linear-programming supervisory HVAC controller (CDC’26).
 - Co-developed an indirect data-driven predictive control method (TPC) built on a state-space predictor (under review).
 - Derived a multi-zone model of how cross-zone heat transfer inflates measured HVAC savings (ACC’26; under review).
 - Co-developed active current-limiting control for home appliances that avoids panel upgrades (CISBAT’25, ACEEE’26).
 - Reviewed 100+ field demonstrations of MPC and reinforcement learning for HVAC (Applied Energy 2025).
- UBC**, Graduate Research Assistant, TMMF Lab & FortisBC Renewable Gas, *BC, Canada* 05/2021–08/2023
- Ran CFD simulations of hydrogen blending into natural gas pipelines; built ideal- and real-gas mixture models in C. 
 - Delivered 1 journal paper, 3 conference papers, an MS thesis, and 2 awards on a project that seeded the UBC H₂Lab. 

INDUSTRY EXPERIENCE

- Purdue Energy & Utilities**, Graduate Researcher, Energy Systems, *West Lafayette, IN* 02/2026–Present
- Leading modeling and data pipelines for a pre-cooling pilot to shift demand off-peak on a 40,320-ton chilled-water system.

- Quantified ~\$41k/yr savings and a statistically significant 8% lower approach from a two-plant air/dirt-separator retrofit.
- Delivered a nine-year energy-use assessment of 21 renovations to Purdue leadership for the Giant Leaps Master Plan.

Tesla, Inc., Applied Machine Learning Intern, Drive Unit, *Sparks, NV*

05/2025–08/2025

- Built MySQL pipelines, engineered features, and trained ML models (XGBoost, random forests, neural nets) with R^2 0.85.
- Helped deploy a thermal imaging system for drive-unit inspection, replacing destructive laboratory testing.
- Delivered ML-driven quality gains projected to cut testing costs by over \$1M/year; offered a three-month extension.

TEACHING EXPERIENCE

Purdue University, Graduate Teaching Assistant (PhD)

- ME597: Distributed Energy Resources (DERs)  *Spring'26*
 - Supported 52 students on modeling and control of DERs such as batteries, solar PV, EVs, and HVAC.
 - Assisted with homework, coding modules, and office hours covering DER concepts.
 - Guided semester projects on DER topics, including building controls and data center cooling.

University of British Columbia, Graduate Teaching Assistant (MS)

- Engineering Analysis I *Fall'21, Spring'22, Fall'22*
- Fluid Mechanics II Lab *Spring'22, Spring'23*

LEADERSHIP & MENTORSHIP

Research Mentor, Purdue University

2024–Present

- Mentored three master's students and one undergraduate on research planning and technical writing.
- Collaborated on their research and co-authored the resulting publications.

Disability Resource Centre Invigilator, University of British Columbia

2021–2023

- Invigilated accommodated exams, supporting students with approved testing accommodations.

ACADEMIC SERVICE

Conference and workshop organizing:

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| July 2026 | Student Chair, Intelligent Building Operations (IBO) Workshop  , <i>West Lafayette, IN</i> |
| July 2024 | Student Chair, Intelligent Building Operations (IBO) Workshop, <i>West Lafayette, IN</i> |

Society memberships: ASHRAE, ACM, IEEE, CSME

Reviewing: IEEE Transactions on Control Systems Technology; IEEE Conference on Decision and Control; Building and Environment; Journal of Building Performance Simulation; ACEEE Summer Study on Energy Efficiency in Buildings; Herrick Conferences

SELECT COURSES

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| • ECE570: Artificial Intelligence | • STAT511: Statistical Methods |
| • IE690: Reinforcement Learning & Control (RLC) | • ME597: Industrial Internet of Things (IIoT) |
| • AAE568: Applied Optimal Control & Estimation | • ME597: Distributed Energy Resources (DERs) |
| • AAE 561: Intro to Convex Optimization | • ME511: Analysis of Thermal Systems |

SKILLS

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| Languages: | Python, MATLAB, C, SQL |
| Optimization & ML: | CVXPY, Gurobi, TensorFlow, PyTorch, scikit-learn, XGBoost, Stable-Baselines3 |
| Simulation & Testbeds: | BOPTEST, Gymnasium, OCHRE |
| Systems & Tools: | InfluxDB, Grafana, Git, Linux, L ^A T _E X, Docker |